

Jacob Horne

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EDUCATION

University of California, Irvine

June 2027

Bachelor of Science in Computer Science; Campuswide Honors Collegium

GPA: 3.9

Coursework: Object-Oriented Programming (C/C++), Data Structures & Algorithms, Databases, Computer Architecture

EXPERIENCE

Capital One

Summer 2026

Software Engineering Intern (Incoming)

New York, NY

Caretech@UCI

Oct 2025 – Present

Software Developer

Irvine, CA

- Processed and prepared **28K+** food images using Python, OpenCV, and PyTorch, building preprocessing pipelines for normalization, augmentation, class balancing, and dataset standardization across **270+** diverse food categories.
- Trained PyTorch-based object detection models using Roboflow and Colab, applying data ablation and augmentation techniques (brightness adjustment, CLAHE, gamma reduction) to improve detection accuracy and model generalization.
- Achieved an **81%** classification accuracy and currently developing a web-based interface for real-time food recognition

Commit the Change

Oct 2025 – Present

Full-Stack Developer

Irvine, CA

- Built a scheduling/quota-management platform with React and TypeScript supporting **23,000+** patients across clinics
- Processed **5,000+** requests/week by developing secure Node.js/Express REST APIs for auth and workflow automation
- Improved staff visibility by implementing modular Chakra UI and Zustand interfaces backed by PostgreSQL data layer
- Reduced booking conflicts and manual scheduling workload by **30%** through real-time tracking appointment system

UCI Digital Learning Lab

Dec 2025 – Present

Research Assistant

Irvine, CA

- Analyzed LLM output datasets using NumPy, Pandas, and Matplotlib for token log-probs, behavior and confidence
- Built Python-based multi-agent evaluation pipelines using LangGraph, implementing critic, advocate, and judge agents to orchestrate rubric-based reasoning, debate workflows, and scoring decisions across essay evaluation pipelines
- Co-authored [paper](#) accepted to **ICLR 2026** LLM Workshop on multi-agent evaluation and token-level LLM confidence

SENS Psychology

Sep 2025 – Jan 2026

Software Development Intern

Arlington, VA

- Built an internal workflow system with Next.js and TypeScript, supporting **30+** staff and **1,200+** monthly interactions
- Reduced patient intake time by **50%** by replacing manual workflows with Node.js/React intake, ticketing, call-routing, and Microsoft Azure notifications backed by PostgreSQL, cutting manual email work **60%** and improving task ownership
- Modeled messaging and task lifecycle in Prisma/Zenstack to support scalable workflow automation and orchestration

PROJECTS

Agonus (Blockchain@UCI) | LangChain, FastAPI, Next.js, Solidity, PostgreSQL

Oct 2025 – Feb 2025

- Built a decentralized AI trading platform where autonomous agents compete in crypto tournaments with on-chain betting, real-time performance tracking with leaderboard, tokenized reward incentives, and a live tournament dashboard UI
- Architected LangChain ReAct agents (GPT-4o-mini) with custom tools for market data, trade execution, planning, and memory; designed Celery/Redis orchestration with scheduling, locking, crash recovery to manage **10+** concurrent agents
- Developed full-stack infrastructure including Solidity betting contracts (OP Sepolia), Uniswap V3 execution, and a Next.js frontend with Wagmi/Viem wallet authentication, transaction signing, and on-chain state synchronization

Prop.Intel - AI Property Risk Platform | Next.js, TypeScript, Python, Ridge Regression

Feb 2026 – Mar 2026

- Built full-stack AI property risk platform generating real-time reports for U.S. properties by integrating **ATTOM** ownership data, **FEMA National Risk Index** datasets, and multi-API pipelines with parallel fetching and caching
- Developed **ridge regression model** across **3,100+** counties and **8** hazard types to quantify disaster exposure and estimate per-property annual loss using normalized county-level risk data and weighted hazard scoring across categories
- Engineered interactive dashboards including ownership timelines, hazard breakdown charts, and AI-generated summaries

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, SQL, Java, Solidity, C#, HTML/CSS, R, MIPS Assembly

Frameworks/Libraries: React, FastAPI, Next.js, Node.js, Express.js, Tailwind CSS, PyTorch, TensorFlow, LangChain

Databases/Tools: PostgreSQL, Git, Docker, Supabase, Prisma, Zenstack, MySQL, Redis, Celery, AWS, Microsoft Azure